

2024 AMC 10B Problem 8 - Solution

Review the full statement and step-by-step solution for 2024 AMC 10B Problem 8. Great practice for AMC 10, AMC 12, AIME, and other math contests

2024 AMC 10B Problem 8

Problem:

Let N be the product of all the positive integer divisors of 42. What is the units digit of N ?

Answer Choices:

- A. 0
- B. 2
- C. 4
- D. 6
- E. 8

Solution:

The prime factorization of 42 is $2 \cdot 3 \cdot 7$. A divisor of 42 is determined by choosing whether or not to include each prime as a factor of the divisor, so there are $2^3 = 8$ divisors. Furthermore, the divisors can be formed into 4 pairs each of whose product is 42 : $1 \cdot 42$, $2 \cdot 21$, $3 \cdot 14$, and $6 \cdot 7$. Therefore $N = 42^4$. The value of the requested units digit of N is the same as the units digit of 2^4 , which is (D) 6.

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